

Method	AP	AUC	Private LB
A Proposed	0.83	0.86	2.46234
B Late-Window	0.66	0.69	2.43530
C Mixed w/o balance	0.78	0.60	2.35424

Table 1. Comparisons on the Nexar validation split. “AP” and “AUC” are computed on the held-out Nexar validation set, with AP reported as the standard average precision in $[0, 1]$. “Private LB” reports the corresponding submission’s score on the official Kaggle *private* leaderboard of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition, where higher is better.

retain the full 5 s input plus a per-frame head.

Cross-Dataset Mixed Training without Class Balancing. Motivated by the scarcity of accident clips in Nexar, we augment training with a second public accident dataset [22] and use the union as the new training set. Crucially, in this experiment we *bypass the class-balancing step of Sec. 2.2*: every positive clip from both sources is kept, and only Nexar’s purely-negative videos are sub-sampled. The resulting training set contains substantially more positives than negatives. The model trained under this regime degenerates into an “always-positive” predictor: on validation it still attains $AP = 0.78$ (Tab. 1, row C), because AP on an imbalanced set rewards any model that tends to predict the majority class, but the corresponding $AUC = 0.60$ —close to the 0.5 chance level—reveals that the model has essentially lost the ability to rank positives above negatives. The private-leaderboard score drops to 2.35, a full -0.108 gap below our final submission. This is a textbook consequence of class imbalance at the dataset level: when the marginal $P(y = 1)$ in training is pushed close to one, the cross-entropy minimiser is the constant-positive predictor, and the time-weighting term in the Exp-Loss—which only re-scales the *positive* loss—provides no counterweight. It is for exactly this reason that we make dataset-construction-time class balancing a first-class component of our pipeline rather than a tunable hyper-parameter.

4. Conclusion

We presented a zero-shot traffic accident anticipation framework that couples a VideoMAE-v2 sliding-window classifier with class-balanced dataset construction and a training-free test-time domain adaptation module. Ablation studies confirm that both full-context per-frame modelling and dataset-level class balancing are essential to avoid performance collapse. The system achieves 2nd place in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1].

References

[1] AUTOPILOT-COG. Zero-shot accident anticipation.

- <https://kaggle.com/competitions/zero-shot-taa>, 2026. Kaggle. 1, 2, 3, 4
- [2] Wentao Bao, Qi Yu, and Yu Kong. Uncertainty-based traffic accident anticipation with spatio-temporal relational learning. In *Proceedings of the 28th ACM International Conference on Multimedia*, pages 2682–2690, 2020. 1, 2
- [3] Jianwu Fang, Dingxin Yan, Jiahuan Qiao, Jianru Xue, and Hongkai Yu. Dada: Driver attention prediction in driving accident scenarios. *IEEE transactions on intelligent transportation systems*, 23(6):4959–4971, 2021. 1
- [4] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022. 2
- [5] Muhammad Monjurul Karim, Yu Li, Ruwen Qin, and Zhaozheng Yin. A dynamic spatial-temporal attention network for early anticipation of traffic accidents. *IEEE Transactions on Intelligent Transportation Systems*, 23(7):9590–9600, 2022. 1
- [6] Dacheng Liao, Mengshi Qi, Liang Liu, and Huadong Ma. Improving batch normalization with test-time adaptation for robust object detection in self-driving. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 6925–6933, 2026. 1
- [7] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 3
- [8] Changsheng Lv, Mengshi Qi, Liang Liu, and Huadong Ma. T2sg: Traffic topology scene graph for topology reasoning in autonomous driving. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 17197–17206, 2025. 1
- [9] Daniel C. Moura, Shizhan Zhu, and Orly Zvitia. Nexar dashcam crash prediction challenge. <https://kaggle.com/competitions/nexar-collision-prediction>, 2025. Kaggle. 2, 3
- [10] Mengshi Qi, Changsheng Lv, and Huadong Ma. Robust disentangled counterfactual learning for physical audiovisual commonsense reasoning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(3):2514–2527, 2026. 1
- [11] Mengshi Qi, Jiaxuan Peng, Xianlin Zhang, and Huadong Ma. Towards balanced multi-modal learning in 3d human pose estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21231–21241, 2026. 1
- [12] Mengshi Qi, Jie Qin, Yi Yang, Yunhong Wang, and Jiebo Luo. Semantics-aware spatial-temporal binaries for cross-modal video retrieval. *IEEE Transactions on Image Processing*, 30:2989–3004, 2021. 1
- [13] Mengshi Qi, Jie Qin, Xiantong Zhen, Di Huang, Yi Yang, and Jiebo Luo. Few-shot ensemble learning for video classification with slowfast memory networks. In *Proceedings of the 28th ACM international conference on multimedia*, pages 3007–3015, 2020. 1
- [14] Mengshi Qi, Yunhong Wang, Annan Li, and Jiebo Luo. Sports video captioning via attentive motion representation

- and group relationship modeling. *IEEE Transactions on Circuits and Systems for Video Technology*, 30(8):2617–2633, 2019. 1
- [15] Mengshi Qi, Yeteng Wu, Xianlin Zhang, and Huadong Ma. Explainable action form assessment by exploiting multimodal chain-of-thoughts reasoning. *arXiv preprint arXiv:2512.15153*, 2025. 1
 - [16] Mengshi Qi, Hao Ye, Jiaxuan Peng, and Huadong Ma. Action quality assessment via hierarchical pose-guided multi-stage contrastive regression. *IEEE Transactions on Image Processing*, 2025. 1
 - [17] Mengshi Qi, Pengfei Zhu, Xiangtai Li, Xiaoyang Bi, Lu Qi, Huadong Ma, and Ming-Hsuan Yang. Dc-sam: In-context segment anything in images and videos via dual consistency. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(4):4642–4656, 2026. 1
 - [18] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078–10093, 2022. 2
 - [19] Haowen Wang, Zhengping Che, Yufan Yang, Mingyuan Wang, Zhiyuan Xu, Xiuquan Qiao, Mengshi Qi, Feifei Feng, and Jian Tang. Rdgc-gan: Rgb-depth fusion cyclegan for indoor depth completion. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(11):7088–7101, 2024. 1
 - [20] Haowen Wang, Mingyuan Wang, Zhengping Che, Zhiyuan Xu, Xiuquan Qiao, Mengshi Qi, Feifei Feng, and Jian Tang. Rgb-depth fusion gan for indoor depth completion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6209–6218, 2022. 1
 - [21] Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yinan He, Yi Wang, Yali Wang, and Yu Qiao. Videomae v2: Scaling video masked autoencoders with dual masking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14549–14560, 2023. 1, 2
 - [22] Yu Yao, Xizi Wang, Mingze Xu, Zelin Pu, Yuchen Wang, Ella Atkins, and David J Crandall. Dota: Unsupervised detection of traffic anomaly in driving videos. *IEEE transactions on pattern analysis and machine intelligence*, 45(1):444–459, 2022. 1, 4
 - [23] Hao Ye, Mengshi Qi, Zhaohong Liu, Liang Liu, and Huadong Ma. Safedriverag: Towards safe autonomous driving with knowledge graph-based retrieval-augmented generation. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 11170–11178, 2025. 1
 - [24] Wulian Yun, Mengshi Qi, Chuanming Wang, and Huadong Ma. Weakly-supervised temporal action localization by inferring salient snippet-feature. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 6908–6916, 2024. 1
 - [25] Pengfei Zhu, Mengshi Qi, Xia Li, Weijian Li, and Huadong Ma. Unsupervised self-driving attention prediction via uncertainty mining and knowledge embedding. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 8558–8568, 2023. 1